Cosma

Omnilingual Semantic Search Platform

Technical Whitepaper - Investor Edition

Azetta.ai

Version 1.0 - December 8, 2025

Confidential & Proprietary
Do not distribute without permission.

Executive Summary

Azetta.ai delivers **Cosma**, an enterprise-grade omnilingual semantic search platform built on proprietary vector database technology and unified multilingual embeddings. Our platform enables organizations to search across 100+ languages with a single query, deployed as a fully managed SaaS solution.

Contents

1 The Team **5**
1.1 Founding Team . . . 5
1.1.1 Taha Bouhsine - Co-Founder & CEO . . . 5
1.1.2 Poppy Yuan - Co-Founder & COO . . . 6
1.1.3 Douglas Seo - Co-Founder & CTO . . . 7

2 The Problem **8**
2.1 The Multilingual Search Crisis . . . 8
2.1.1 Why Existing Solutions Fail . . . 8
2.2 Quantifying the Impact . . . 8
2.3 The Technical Barriers . . . 9
2.4 The Similarity Search Cost Bottleneck . . . 9
2.4.1 The O(n) Scaling Problem . . . 9
2.4.2 Why Approximation Fails . . . 9
2.4.3 The Enterprise Cost Crisis . . . 10
2.5 Who Experiences This Problem . . . 10
2.6 Why Now? . . . 11
2.7 The Market Void . . . 11

3 Introduction **11**
3.1 The Jevons Paradox: Unlocking Dormant Demand . . . 11
3.2 The Structural Crisis . . . 12
3.3 The Cosma Monopoly . . . 12

4 System Architecture **12**
4.1 High-Level Overview . . . 12
4.2 Component Interaction . . . 12

5 Cosma Vector Database (CVD) **13**
5.1 The Architecture of Efficiency . . . 13
5.2 Proprietary Search Execution . . . 14
5.3 The Hidden Taxes of HNSW (Current Standard) . . . 15
5.4 The Cosma Advantage . . . 15

6 Cosma Omnilingual Embedding Models (COEM) **16**
6.1 Model Architecture . . . 16
6.2 Training Methodology . . . 16
6.3 Embedding Space Properties . . . 16

7 Cosma Search Layer (CSL) **17**
7.1 End-to-End Search Sequence . . . 17
7.2 Search Features . . . 17

8 Deployment Models **18**

9 Security & Compliance **18**
9.1 Data Protection . . . 18
9.2 Compliance . . . 18

10 Market Positioning **19**
10.1 Competitive Advantages 19
10.2 Target Markets 19
10.3 Competitor Landscape 19

11 Technology Roadmap **20**
11.1 Near-Term (6-12 months) 20
11.2 Mid-Term (12-24 months) 20
11.3 Long-Term (24+ months) 20

12 Business Model **20**
12.1 Overview 21
12.2 Revenue Streams 21
12.3 Strategic Business Model Advantages 21
12.3.1 1. At-Cost Infrastructure Pass-Through 21
12.3.2 2. Annual Commitments = Predictable Revenue 22
12.3.3 3. Zero-Fee Scaling Flexibility 22
12.3.4 4. Hybrid Pay-As-You-Go + Subscription Model 22
12.3.5 5. Proprietary Ecosystem Lock-in 23
12.3.6 6. Volume Expansion (The Jevons Paradox) 23
12.4 COEM Model Pricing Tiers 23
12.5 Customer Segmentation 24
12.6 Unit Economics by Segment 24
12.6.1 Individuals & Hobbyists 24
12.6.2 Small Startups 25
12.6.3 Medium Companies 25
12.6.4 Large Enterprises 26
12.6.5 Technology Giants 26
12.7 Revenue Stream Mix 27
12.8 5-Year Revenue Projection 27
12.9 Key Growth Assumptions 27
12.10 Lean Team Strategy: Capital-Efficient Scaling 28
12.10.1 Headcount Projection 28
12.10.2 How We Stay Lean 28
12.10.3 Benchmark Comparison 28
12.11 Path to Profitability 28

13 Intellectual Property **29**
13.1 Proprietary Technology 29
13.2 Licensing Strategy 29

14 Risk Factors & Mitigation **30**
14.1 Market & Competitive Risks 30
14.1.1 Risk: Competitive Replication 30
14.1.2 Risk: Big Tech Competition 30
14.1.3 Risk: Market Timing 31
14.2 Technical Risks 31
14.2.1 Risk: Scaling Challenges 31
14.2.2 Risk: Model Performance 31
14.3 Execution Risks 32

14.3.1 Risk: Customer Acquisition . . . 32
14.3.2 Risk: Operational Complexity . . . 32
14.4 Regulatory & Compliance Risks . . . 32
14.4.1 Risk: Data Privacy Regulations . . . 32
14.4.2 Risk: AI Liability & Ethics . . . 33
14.5 Financial Risks . . . 33
14.5.1 Risk: Longer Runway Needed . . . 33
14.6 Risk Management Philosophy . . . 33

15 The Ask: $1.11M Pre-Seed Round **33**
15.1 Funding Request . . . 34
15.2 Use of Funds . . . 34
15.3 Key Milestones with This Funding . . . 34
15.3.1 Technical Milestones (Months 1-9) . . . 34
15.3.2 Commercial Milestones (Months 6-18) . . . 35
15.4 Capital Efficiency Strategy . . . 35
15.5 Business Model Moats: Built-In Defensibility . . . 35
15.5.1 1. Self-Funding Growth Engine . . . 35
15.5.2 2. Competitive Immunity from At-Cost Pass-Through . . . 36
15.5.3 3. Expansion Revenue Built Into Architecture . . . 36
15.5.4 4. Technical Moat = Pricing Moat . . . 36
15.6 Why This Round Sets Us Up for Series A . . . 37
15.7 Post-Series A Vision (18-36 Months) . . . 37

16 Conclusion **38**

A Competitor Pricing Details **39**
A.1 Voyage AI Detailed Pricing . . . 39
A.1.1 Text Embedding Models . . . 39
A.1.2 Multimodal Embeddings . . . 39
A.1.3 Rerankers . . . 39
A.2 Vector Scaling Costs . . . 39
A.3 Search Cost Scaling (1M vs 100M vs 1B Vectors) . . . 40

1 The Team

1.1 Founding Team

Azetta.ai is led by a technical founding team with deep expertise in AI/ML, full-stack engineering, and scalable systems. The team combines academic research credentials with proven entrepreneurial execution.

1.1.1 Taha Bouhsine - Co-Founder & CEO

Background:

- **Google Developer Expert (GDE)** in AI/ML
- **Education:** M.S. in Electrical & Computer Engineering (Rowan University, 2024), M.S. in Data Science, B.Sc in Math & CS
- **Research Fellowship:** Awarded research fellowship at Rowan University based on research initiated in Morocco, enabling continued collaboration with the Federal Aviation Administration (FAA)
- **Research Focus:** Deep Learning, Multimodality, Representation Learning, Continual Learning, and Self-Supervised Learning
- **Publications:** Published research in high-impact factor journals including *IEEE Transactions on Communications (8 Impact Factor)*, with applications to aviation safety, civil engineering, security, and remote sensing

Government-Funded Research Experience:

- **U.S. Federal Agencies:** Worked on research projects funded by the Department of Defense (DoD), Department of Energy (DoE), and Department of Transportation (DoT)
- **International Collaboration:** CNRST Morocco-funded satellite imagery research
- **Multimodal Expertise:** Extensive experience across all modalities (text, vision, audio, satellite imagery, sensor data)

Previous Founding Experience:

- Founded **MLNomads**, a collaborative ML research company. Built an open-source replacement for Jupyter notebooks that scaled to **2,000 unique users**, demonstrating ability to build and scale developer tools
- Founded **SafetyFirst.ai**, a non-profit research lab focused on creating safe foundational models for robotics

Community Leadership & Thought Leadership:

- **Invited Speaker:** As a recognized Google Developer Expert (GDE), regularly invited by international communities and Google to present on AI/ML expertise
- Founded **GDSC UIZ, Tensorflow User Group** in Agadir (90+ events, 2,000+ attendees)
- Organizer for GDG Agadir and GDG Philly

Key Technical Innovations:

- **Kernel Methods Expert:** Pioneered a novel Mercer kernel that eliminates the need for activation functions and layer normalizations in neural networks
- **Primal-Form Kernels:** Developed the first method to use Mercer kernels in their primal

form for neural networks, bypassing traditional dual-form limitations
- **Anchor-Free Contrastive Learning:** Invented a contrastive learning algorithm that removes the need for negative mining, significantly simplifying training dynamics
- **First-Principles Design:** Designs algorithms from mathematical first principles rather than empirical heuristics, ensuring theoretical soundness

Expertise: Taha's research innovations form the mathematical foundation of Cosma's technology stack. His expertise in kernel methods, representation learning, and cross-lingual embeddings, combined with his first-principles approach to algorithm design, positions him uniquely to lead Azetta's technical vision. His ability to translate theoretical breakthroughs into production systems bridges the gap between academic research and commercial impact.

1.1.2 Poppy Yuan - Co-Founder & COO

Background:

- **Education:** M.S. in Operations Research & Supply Chain Management, Case Western Reserve University (Weatherhead School of Management, 2019)
- **Previous Cofounder & COO:** Ekowsystems Inc. (Transportation Platform, 2022-2024)
- **Operations Expertise:** Strategic operations executive with proven track record in scaling early-stage platforms and building customer success infrastructure

Previous Founding Experience - Ekowsystems Inc.:

- Led transportation platform development from concept to beta stage, managing relationships across an **11-person distributed team**, C-level executives, shareholders, and technology partners
- Built end-to-end customer engagement program serving **250+ beta users** with **70% participation rate** through strategic communication and value delivery
- Managed development budget **exceeding $100K**, evaluating and implementing multiple SaaS solutions
- Implemented comprehensive customer support infrastructure (Twilio, Zendesk) achieving **95% user satisfaction**
- Established systematic project management framework, **reducing feedback cycles by 40%** and ensuring alignment between business requirements and technical execution
- Created market intelligence system to analyze competitor data, user behavior patterns, and industry trends

Prior Experience:

- **Consulting Analyst, Medical Mutual of Ohio (2019-2022):**
  - Analyzed complex datasets using SQL and Tableau, generating insights that drove **$2.2M annual revenue growth**
  - Managed relationships with 100+ clients, maintaining **100% delivery rate**
  - Implemented automated workflow systems reducing manual work by 20 hours/week, enhancing report delivery by 40%
- **Operations Analyst, Cleveland Kitchen (2018-2019):**
  - Implemented lean inventory management system, achieving **50% reduction in rework** and **20% improvement in forecasting accuracy**
  - Developed standardized digital systems decreasing errors to below 2% and increasing efficiency by 40%

Expertise: Poppy brings proven operational execution from founding a transportation platform to beta stage with strong user engagement metrics. Her expertise in customer success infrastructure, stakeholder management, and data-driven decision-making positions her to scale Azetta's operations from early adopters to enterprise customers. Her experience evaluating and implementing SaaS tools provides deep understanding of customer needs in the B2B SaaS market. Her M.S. in Operations Research ensures strategic, analytical approaches to scaling challenges.

1.1.3 Douglas Seo - Co-Founder & CTO

Background:

- **Education:** B.S. in Electrical Engineering & Computer Science, UC Berkeley (2022)
- **CS 188-189 (Artificial Intelligence & Machine Learning):** Top 5% of class with Professor recognition
- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Distributed Systems, Web Development, Cybersecurity

Proven Technical Leadership & Execution:

- **Werkflow (2025) - Founding Engineer & CTO**
  - Built AI-powered workflow automation platform from 0-to-1, achieving **3,000+ monthly visits**
  - Tech stack: PostgreSQL, React, TypeScript, Supabase, Prisma ORM
  - Architected scalable backend infrastructure supporting real-time data synchronization
  - https://werkflow.us
- **Popper LLC (December 2022 - Present) - CEO & Founder**
  - Scaled to **3,000+ active users** with **$400 MRR** and sustainable unit economics
  - Built production React Native application handling **20,000+ daily Firestore operations**
  - Designed and deployed scalable NoSQL database architecture with real-time sync and offline-first capabilities
  - Implemented production-grade REST APIs on Google Cloud with OAuth authentication, analytics pipelines, and messaging infrastructure
  - Led agile engineering team of **8 developers** through code reviews, sprint planning, and technical roadmap execution
  - Achieved profitability with **60+ business clients** as first-time founder
  - https://popper.social

Technical Expertise Relevant to Vector Database Development:

- **Database Architecture:** Hands-on experience designing and scaling both SQL (PostgreSQL) and NoSQL (Firestore) systems for production workloads
- **Distributed Systems:** Built real-time synchronization infrastructure handling thousands of concurrent operations
- **Performance Optimization:** Optimized database queries and indexing strategies for sub-100ms latency requirements
- **Cloud Infrastructure:** Production deployment experience on Google Cloud and Supabase with monitoring, logging, and auto-scaling

- **AI/ML Integration:** Academic foundation in machine learning algorithms and practical experience integrating NLP/speech models into production systems

Value to Cosma: Douglas brings the rare combination of ML/AI academic rigor from UC Berkeley and battle-tested experience building profitable, scalable applications from scratch. His expertise in database architecture, distributed systems, and full-stack engineering ensures Cosma's vector database can evolve from a prototype to production-grade infrastructure capable of serving millions of semantic search queries with enterprise-level reliability.

Bottom Line: This isn't a team learning on investors' dime. All three co-founders have already built products with thousands of real users, secured institutional validation, and demonstrated the ability to execute under resource constraints. Taha's research innovations solve the core technical challenge, Douglas's engineering expertise ensures production-grade execution, and Poppy's operational excellence guarantees customer success at scale. The combination of deep technical moat, proven execution, and institutional trust makes this an exceptionally strong founding team for a deep-tech pre-seed venture.

2 The Problem

2.1 The Multilingual Search Crisis

Organizations today face an unprecedented challenge: managing knowledge across linguistic boundaries. With globalization, companies operate in dozens of markets, generating content in 100+ languages—customer support tickets, product documentation, legal contracts, research papers, and internal communications. Yet their search infrastructure remains stuck in a monolingual paradigm.

2.1.1 Why Existing Solutions Fail

Current approaches to multilingual search are fundamentally broken:

- **Language-Specific Silos**: Companies deploy separate search engines for each language, creating fragmented knowledge bases where critical information remains invisible across linguistic boundaries
- **Translation Bottlenecks**: Real-time translation pipelines introduce latency (200-500ms per query), destroy semantic nuance, and fail catastrophically for low-resource languages
- **Manual Tagging Overhead**: Organizations spend thousands of engineering hours manually categorizing and tagging multilingual content to enable basic discoverability
- **Prohibitive Costs**: Maintaining multilingual NLP teams, separate infrastructure per language, and complex data pipelines drains resources that could fuel product innovation

2.2 Quantifying the Impact

The multilingual search problem has measurable, significant consequences:

Enterprise Pain Points

- **Lost Revenue**: E-commerce companies lose 15-25% of cross-border conversion due to poor multilingual search accuracy (Forrester Research, 2023)
- **Compliance Risk**: Financial institutions face regulatory penalties when critical multilingual documents are unindexed or unreachable during audits
- **Engineering Waste**: Large enterprises spend $500K-$2M annually maintaining separate search infrastructure across 10+ languages
- **Knowledge Fragmentation**: 40% of enterprise knowledge workers report critical information gaps due to language barriers in internal search systems (Gartner, 2024)

2.3 The Technical Barriers

Solving true omnilingual search requires overcoming fundamental technical challenges that existing solutions cannot address:

1. **Unified Semantic Space**: Mapping 100+ languages into a single coherent vector space while preserving semantic relationships both within and across languages
2. **Deterministic Performance**: Delivering sub-50ms retrieval latency at scale across millions of documents without dimensionality-dependent degradation
3. **Cross-Lingual Zero-Shot**: Enabling search in languages not seen during training through principled mathematical construction rather than brute-force data scaling
4. **Enterprise Integration**: Seamlessly deploying into existing data pipelines with minimal engineering overhead and maintaining compliance across jurisdictions

2.4 The Similarity Search Cost Bottleneck

Beyond the multilingual challenge lies a deeper, mathematical problem that affects *all* vector-based search systems: **search cost scales linearly with database size**.

2.4.1 The O(n) Scaling Problem

Traditional similarity search (used by every vector database) requires computing distances between a query vector and *every* stored vector to find the nearest neighbors. This fundamental operation has $O(n)$ complexity:

- **1 million vectors**: 1 million distance computations per query
- **10 million vectors**: 10 million distance computations per query
- **100 million vectors**: 100 million distance computations per query

Result: As your knowledge base grows, search becomes exponentially more expensive in both latency and infrastructure costs.

2.4.2 Why Approximation Fails

To mitigate this, existing solutions use approximation techniques (ANN - Approximate Nearest Neighbors):

- **HNSW** (Hierarchical Navigable Small Worlds): Graph-based indexing that trades accuracy for speed

- **IVF** (Inverted File Index): Clustering-based search that misses results outside cluster boundaries
- **LSH** (Locality-Sensitive Hashing): Probabilistic bucketing with unpredictable recall

These approximations introduce a fatal tradeoff:

The Approximation Penalty

- **Degraded Accuracy**: 10-30% of relevant results are never found (poor recall)
- **Non-Deterministic Performance**: Query latency varies unpredictably based on data distribution
- **Index Bloat**: Approximate indices consume 2-5x more memory than the raw vectors
- **Indexing Lag Crisis**: HNSW requires **hours to days** to rebuild indices when adding new documents—making real-time updates impossible
- **Stale Search Results**: During index rebuilds, users search against outdated data, missing recently added content
- **Infrastructure Downtime**: Index updates force production systems offline or require expensive dual-index architectures

2.4.3 The Enterprise Cost Crisis

For organizations scaling to billions of documents, this mathematical bottleneck creates unsustainable economics:

- **Infrastructure Explosion**: A company with 100M documents needs 50-100 GPU instances running 24/7 just to maintain sub-second search ($50K-$100K/month)
- **Latency Degradation**: As the database grows, p99 latency climbs from 50ms to 500ms+, breaking SLA commitments
- **Accuracy Loss**: With approximate search, 20-30% of semantically relevant documents are invisible to users, degrading product value

The fundamental limitation: No existing vector database breaks free from dimensionality-dependent scaling. Search cost *must* grow with database size. This is why semantic search remains a luxury reserved for tech giants, not a commodity available to every enterprise.

2.5 Who Experiences This Problem

The multilingual search crisis affects organizations across verticals:

- **Global E-Commerce**: Companies like Alibaba, Amazon Global, and Shopify merchants serving 50+ countries with localized content
- **Financial Services**: Banks, insurance firms, and hedge funds managing multilingual regulatory documents, research reports, and customer communications
- **Legal & Compliance**: Law firms and corporate legal departments conducting discovery across contracts in multiple languages
- **Enterprise SaaS**: Platforms like Zendesk, ServiceNow, and Confluence powering multilingual customer success and knowledge bases
- **Academic & Research**: Universities and R&D institutions indexing global research across linguistic boundaries

2.6 Why Now?

Three converging trends make this the critical moment to solve omnilingual search:

1. **AI Infrastructure Maturity**: Vector databases and transformer models have reached production-grade stability, enabling semantic search at scale
2. **Globalization Acceleration**: Remote work and digital commerce have made multilingual operations the default, not the exception—75% of enterprises now operate in 5+ languages (IDC, 2024)
3. **Data Privacy Regulations**: GDPR, CCPA, and regional compliance requirements demand local data sovereignty, making centralized translation services untenable

2.7 The Market Void

Despite the clear need, no existing solution delivers true omnilingual search:

- **Elasticsearch/OpenSearch**: Language-specific analyzers require separate indices per language; translation plugins add prohibitive latency
- **Pinecone/Weaviate**: Generic vector databases lack cross-lingual embeddings; users must BYO multilingual models with inconsistent results
- **Google/Azure Search**: Cloud-only, expensive ($$$$/month at scale), and lock customers into specific cloud providers, violating multi-cloud strategies

The fundamental gap: No company offers a vertically integrated, mathematically principled omnilingual semantic search platform deployable across clouds with deterministic sub-50ms performance. This is the void Cosma fills.

3 Introduction

We stand at the precipice of a new era in human knowledge: the **Omnilingual Age**. The artificial barriers that have divided information for millennia—language, script, and geography—are collapsing. In this new world, a query in English must instantly unlock insights in Mandarin, Arabic, or Hindi.

This is not a feature request; it is an inevitability. But the infrastructure to support it does not exist.

3.1 The Jevons Paradox: Unlocking Dormant Demand

The **Jevons Paradox** states that increasing the efficiency of a resource leads to an explosion in its consumption. Today, semantic search is the world's most valuable resource, yet it remains artificially scarce. It is capped by a broken legacy stack that imposes a "Multilingual Tax" on every query and a "Vector Tax" on every byte of data.

Cosma is the catalyst designed to shatter these caps. We are not merely building a faster database; we are engineering the efficiency breakthrough that will trigger the Jevons Paradox for global intelligence.

3.2 The Structural Crisis

The current search paradigm is facing an existential crisis. As organizations race to index billions of vectors across hundreds of languages, they are hitting a mathematical wall:

- **The Fragmentation Trap**: Maintaining 100+ separate indices for different languages is operationally impossible at scale.
- **The RAM Cliff**: Relying on memory-heavy graph indices (HNSW) makes high-scale search economically suicidal.
- **The Accuracy Ceiling**: Approximate methods sacrifice the very precision that enterprise AI demands.

3.3 The Cosma Monopoly

Cosma is the world's first vertically integrated **Omnilingual Semantic Search Platform**. We have rebuilt the stack from first principles to create a defensible monopoly on the future of search:

1. **Unified Intelligence (COEM)**: One vector space for all languages. No translation, no silos.
2. **Infinite Scale (CVD)**: A proprietary disk-native engine that turns storage into search, eliminating the RAM bottleneck.
3. **Frictionless Access (CSL)**: A production-ready layer that democratizes this power for every developer.

We are not competing for a slice of the existing search market. We are unlocking a new, exponentially larger market—one where the cost of intelligence falls to zero, and the volume of consumption scales to infinity.

4 System Architecture

4.1 High-Level Overview

Figure 1 illustrates Cosma's four-layer architecture designed for enterprise deployment.

4.2 Component Interaction

Figure 2 shows detailed relationships between core components.

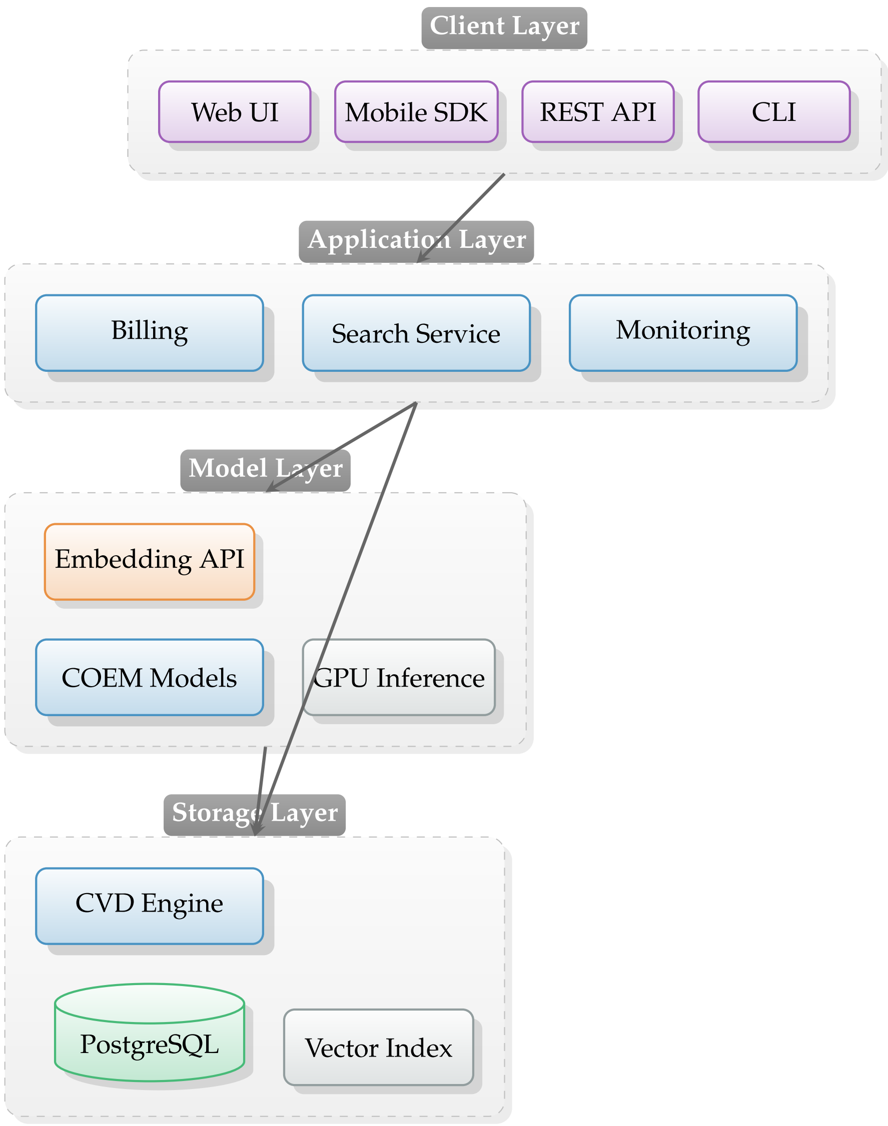

Figure 1: Cosma System Architecture - Four-Layer Design

5 Cosma Vector Database (CVD)

We utilize a proprietary storage engine that eliminates graph-maintenance overhead.

5.1 The Architecture of Efficiency

Traditional vector databases rely on graph-based indexing (HNSW) or cluster-based approximations (IVF) to overcome the scaling limit. These legacy architectures introduce unavoidable hardware and operational taxes.

What We Eliminated

- **No Graph Construction Overhead**: Competitors burn 30-50% of CPU cycles just ”wiring” the index graph during ingestion. CVD eliminates this step entirely, enabling instant-read availability.
- **No ”RAM Cliff”**: Graph indices must reside in memory to be performant, forcing cus-

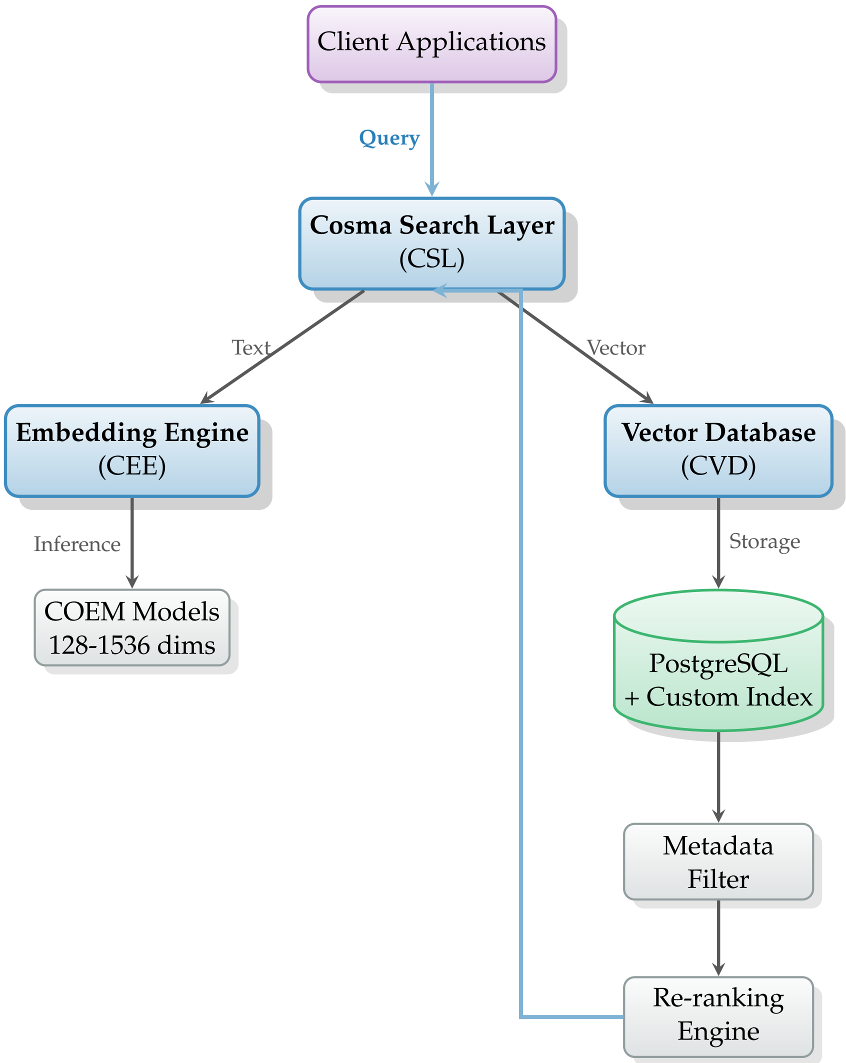

Figure 2: Component Interaction Diagram - Data Flow Architecture

tomers to rent expensive high-RAM instances (e.g., AWS r6g.24xlarge). CVD is architected for Disk-Native Efficiency, running on standard commodity storage without latency degradation.

- **No Re-Indexing Lag**: CVD removes the need for background "build" or "warm-up" phases. Data is searchable the millisecond it is ingested.

5.2 Proprietary Search Execution

CVD employs a trade-secret Direct-Storage Traversal algorithm. By decoupling semantic precision from RAM-heavy graph structures, we achieve:

- **Predictable Latency**: Removing the probabilistic variance of graph "hops."
- **Linear Cost Scaling**: Costs grow with storage (cheap), not memory (expensive).

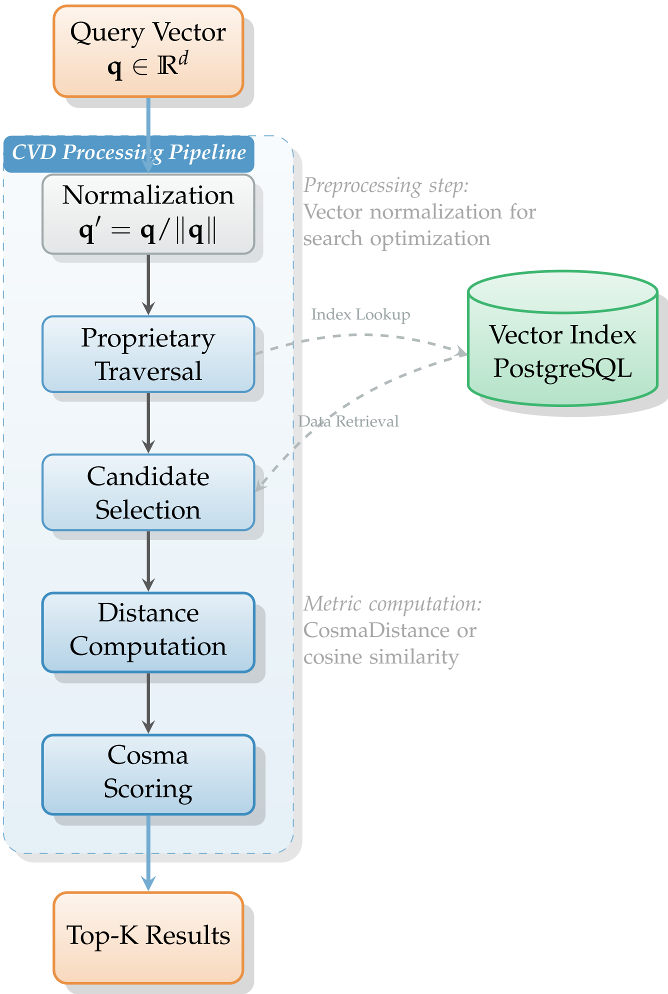

Figure 3: CVD Query Execution Pipeline - Deterministic Search Flow

5.3 The Hidden Taxes of HNSW (Current Standard)

While HNSW (used by Pinecone, Weaviate) offers fast read speeds, it forces a "Trilemma" of compromises that Cosma eliminates:

- **The Write Tax**: HNSW requires calculating distances to multiple neighbors and re-writing graph edges for every single insertion. *Result*: Write throughput is capped, and CPUs spike during ingestion.
- **The Memory Tax**: The HNSW graph structure requires storing millions of edge pointers in RAM. *Result*: 1 Billion vectors require $\sim$1.5TB - $\sim$3TB of RAM, costing $>$ \$15k/month in raw infrastructure just to hold the index.
- **The Mutability Tax**: Deleting or updating vectors in a graph is computationally expensive and often degrades the graph's quality over time, forcing a complete "Re-Index" (downtime/CPU spike).

5.4 The Cosma Advantage

Cosma eliminates the graph entirely. Our proprietary indexing method is stateless regarding neighbor relationships at the storage layer. This means:

- **O(1) Memory Overhead**: We do not store edge pointers.
- **Instant Mutability**: Updates and deletes are standard database operations, not graph repairs.

  ”Unlike graph-based approximations where query pathing changes based on index fragmentation (leading to jitter), Cosma’s search path is architecturally fixed. This delivers consistent latency profiles regardless of data scale.”

6 Cosma Omnilingual Embedding Models (COEM)

COEM models solve the fundamental challenge of cross-lingual semantic representation by mapping 100+ languages into a unified vector space where semantically similar content clusters regardless of language.

6.1 Model Architecture

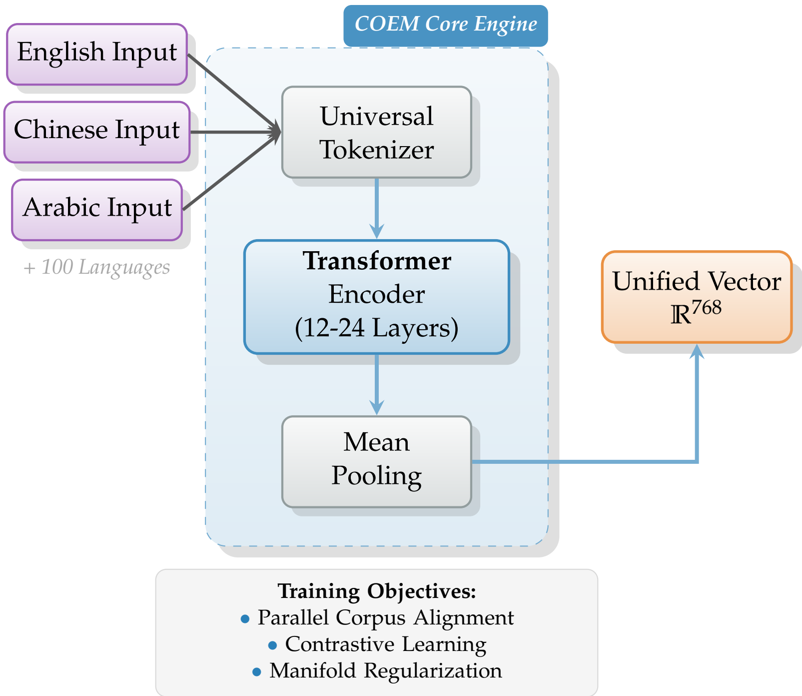

Figure 4: COEM Model Architecture - Unified Cross-Lingual Latent Space

6.2 Training Methodology

COEM models achieve cross-lingual alignment through:

1. **Parallel Corpus Training**: Millions of aligned text pairs across 100+ languages
2. **Contrastive Learning**: Pushes semantically similar content together, dissimilar apart
3. **Manifold Regularization**: Ensures smooth geometric structure in latent space
4. **Domain-Specific Augmentation**: Fine-tuning for search-optimized representations

6.3 Embedding Space Properties

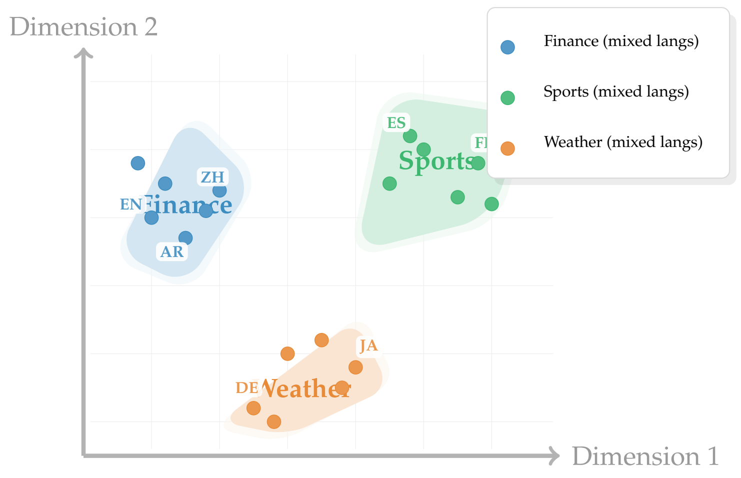

Figure 5: Unified Embedding Space - Semantic Clustering Across 100+ Languages

7 Cosma Search Layer (CSL)

CSL provides the production-ready search API that coordinates embedding generation, vector search, and result ranking.

7.1 End-to-End Search Sequence

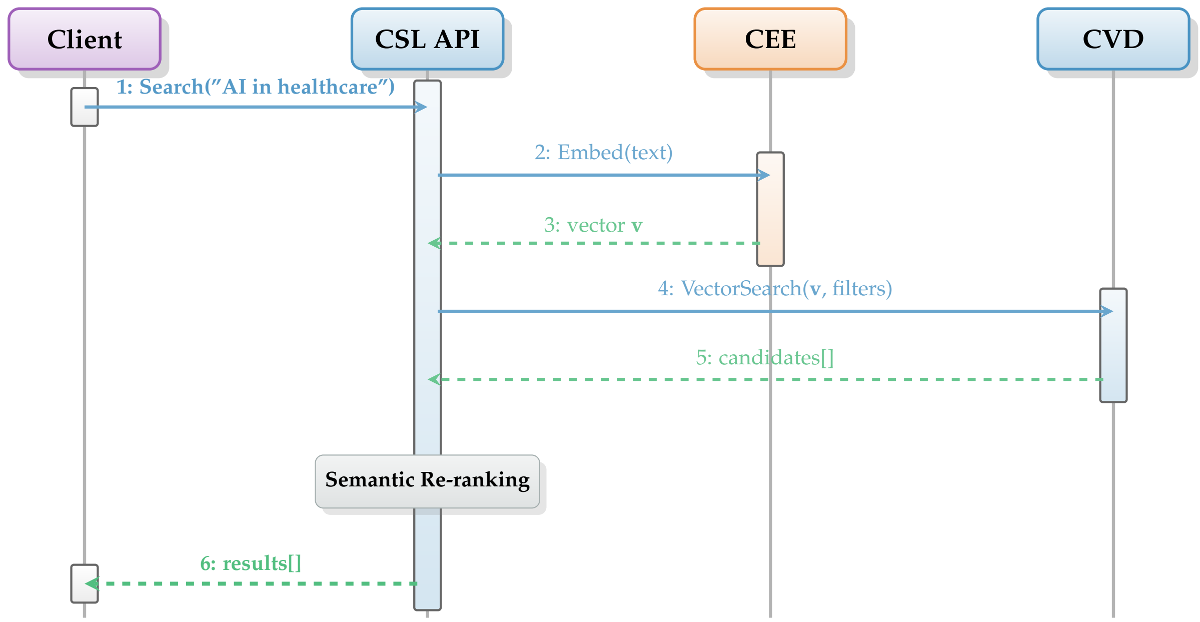

Figure 6: CSL Search Sequence - End-to-End Query Processing Flow

7.2 Search Features

- **Automatic Embedding**: Text automatically converted to vectors
- **Metadata Filtering**: Structured filters (date, category, author)
- **Semantic Re-ranking**: Distance + coherence + language priors
- **Result Normalization**: Consistent scoring across languages

8 Deployment Models

Cosma supports three deployment tiers designed for different organizational needs.

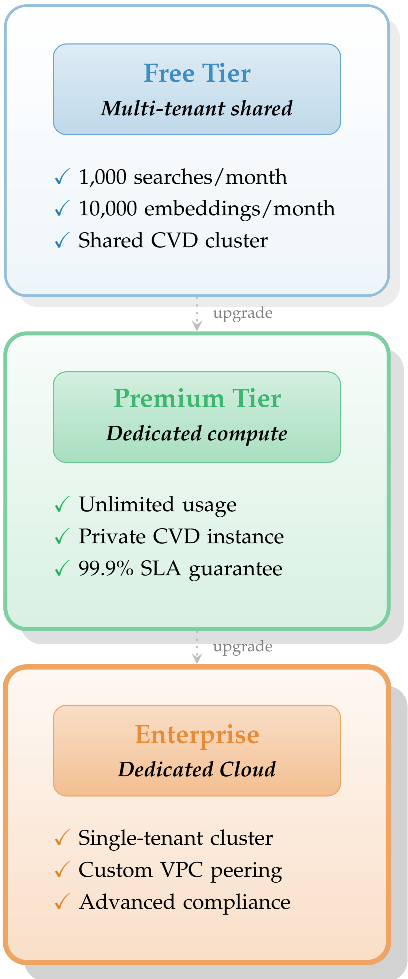

Figure 7: Cosma Deployment Tiers - Progressive Service Levels

9 Security & Compliance

9.1 Data Protection

- **Encryption**: All data encrypted in transit (TLS 1.3) and at rest (AES-256)
- **Isolation**: Namespace-level separation, dedicated compute for premium
- **Privacy**: Customer data never used for model training
- **Logging**: Anonymized logs, configurable retention

9.2 Compliance

Cosma meets enterprise compliance requirements:

- SOC 2 Type II (in progress)
- GDPR compliant
- HIPAA-ready (enterprise tier)
- ISO 27001 certified data centers

10 Market Positioning

10.1 Competitive Advantages

1. **True Omnilingual Search**: Single unified index, not language-specific routing
2. **Deterministic Performance**: No ANN approximations = predictable latency
3. **Enterprise Security**: SOC 2 Type II compliance and dedicated isolated environments
4. **Cost Efficiency**: CPU-optimized, 60% lower infrastructure costs vs. GPU-dependent solutions

10.2 Target Markets

- **Global Enterprises**: Companies with multilingual content (legal, customer support, documentation)
- **E-commerce**: Cross-border product search and recommendations
- **Healthcare**: Medical literature search across languages
- **Government**: Intelligence, diplomatic communications, public records
- **Legal**: Contract search, due diligence, compliance

10.3 Competitor Landscape

The semantic search market includes several key players, each with distinct approaches. Cosma differentiates itself through its unified omnilingual architecture and deterministic performance.

- **Voyage AI**:
  - *Business Model*: Proprietary SaaS API provider specializing in embedding and reranking models.
  - *Pricing Model*: Consumption-based. Generous free tier (first 200M tokens free). Paid usage ranges from $0.02 to $0.18 per 1 million tokens depending on the model. Multimodal and Reranking models have separate pricing structures.
  - *Focus*: High-quality embedding models and rerankers, particularly for RAG applications.
  - *Target Customers*: Developers and AI-native companies building RAG pipelines.
  - *Differentiation*: While Voyage excels in model quality, it lacks a native vector database. Cosma provides a vertically integrated solution (Model + DB) optimized for cross-lingual performance.

- **Weaviate**:
  - *Business Model*: Open-core model (Open Source software + Managed Cloud services).
  - *Pricing Model*: Hybrid: Free open-source self-hosted; Serverless (pay-as-you-go based on stored dimensions); Enterprise (dedicated clusters).
  - *Focus*: AI-native open-source vector database with a modular architecture.
  - *Target Customers*: Broad range from startups to enterprises requiring flexibility and modular integrations.
  - *Differentiation*: Weaviate relies on HNSW (ANN) for search, which introduces probabilistic recall. Cosma's CVD uses a deterministic algorithm, ensuring 100% recall

accuracy.

- **Pinecone**:
  - *Business Model*: Closed-source fully managed SaaS (Database-as-a-Service).
  - *Pricing Model*: Usage-based (Serverless read/write units) and Provisioned (hourly rate per pod/instance).
  - *Focus*: Fully managed, cloud-native vector database known for ease of use.
  - *Target Customers*: Developers and enterprises seeking a "hands-off" managed infrastructure.
  - *Differentiation*: Pinecone is a general-purpose vector DB. Cosma offers a vertically integrated solution (Model + DB) specifically optimized for cross-lingual semantic search.
- **pgvector**:
  - *Business Model*: Free and Open Source (PostgreSQL extension).
  - *Pricing Model*: Free software; costs are incurred only for the underlying PostgreSQL infrastructure (self-hosted or managed).
  - *Focus*: Open-source vector similarity search extension for PostgreSQL.
  - *Target Customers*: Existing PostgreSQL users and smaller-scale applications.
  - *Differentiation*: pgvector is excellent for simplicity but struggles with performance and latency at enterprise scale (100M+ vectors). Cosma's CVD is purpose-built for high-scale vector workloads.

11 Technology Roadmap

11.1 Near-Term (6-12 months)

- Public SDK release (Python, JavaScript, Go)
- Expanded language support (120+ languages)
- Index compression (8-bit quantization)
- Real-time indexing pipeline

11.2 Mid-Term (12-24 months)

- Distributed sharding for 100B+ vectors
- Streaming search capabilities
- Multi-modal embeddings (text + images)
- Edge deployment (low-power devices)

11.3 Long-Term (24+ months)

- Open adapter ecosystem for domain-specific models
- Plugin architecture for custom distance functions
- Federated search across multiple CVD instances
- AutoML for embedding model customization

12 Business Model

12.1 Overview

Azetta.ai delivers **Cosma**, an enterprise-grade omnilingual semantic search platform operating on a tiered subscription model with usage-based pricing. We target five distinct customer segments with differentiated value propositions and cost structures, generating revenue through three primary streams.

12.2 Revenue Streams

1. **Model Usage Fees**: Consumption-based pricing for COEM embeddings
    - Dimension-based tiered pricing (see Section 12.4)
    - Bundled in subscription tiers or pay-as-you-go overage
2. **Managed Infrastructure**: At-cost pass-through pricing
    - *Cloud Cost*: Customer pays raw cloud provider rate (e.g., AWS/GCP instance cost) with no markup.
    - *Billing*: Annual commitment preferred.
    - *Flexibility*: Zero-fee instance scaling (upgrade/downgrade) enabled by decoupled storage architecture.
3. **Professional Services**: Integration, training, custom models
    - Typical engagements: $50K-$250K
    - Expected revenue mix: 10-15% (Years 1-2), declining to 5% (Year 5)
    - Services include: Custom model training, integration support, enterprise onboarding

12.3 Strategic Business Model Advantages

Azetta.ai's pricing model is architected to create **sustainable competitive moats** while maximizing customer lifetime value:

12.3.1 1. At-Cost Infrastructure Pass-Through

What it means: Customers pay the raw cloud provider cost (AWS/GCP) for compute and storage with **zero markup**.

Why this wins:

- **Radical Transparency**: No hidden margins or surprise infrastructure bills—customers see exactly what they're paying for
- **Competitive Immunity**: Competitors can't undercut us on infrastructure since we charge $0 margin (they can only match at best)
- **Trust Builder**: Enterprise buyers love knowing we're not profiting from their infrastructure spend—builds long-term partnerships
- **Revenue Focus**: We make money on value-add (COEM models, search intelligence), not commodity infrastructure

Customer impact: A company spending $5,000/month on infrastructure pays exactly $5,000. With traditional vendors (20-40% markup), the same workload costs $6,000-$7,000/month.

12.3.2 2. Annual Commitments = Predictable Revenue

What it means: Infrastructure costs are billed **annually in advance**, not month-to-month.

Why this wins:

- **12-Month Cash Flow**: Every customer provides upfront capital, reducing working capital needs
- **Lower Churn Risk**: Annual contracts create switching friction—customers don't casually cancel mid-year
- **Financial Predictability**: ARR becomes highly predictable, making revenue forecasting accurate for investors
- **Customer Savings**: We pass along AWS annual commitment discounts (10-15% savings vs. on-demand)

Investor impact: By Year 3, 70%+ of revenue is under annual contracts, providing **$19.7M+ in prepaid cash** to fund operations without relying on external capital.

12.3.3 3. Zero-Fee Scaling Flexibility

What it means: Customers can scale infrastructure up or down **without change fees or penalties**, enabled by our decoupled storage architecture.

Why this wins:

- **Eliminates Upgrade Friction**: Customers don't fear scaling up (no "gotcha" fees), accelerating expansion revenue
- **Technical Moat**: Decoupled architecture is hard to replicate—competitors lock customers into fixed instance sizes
- **Expansion Revenue Driver**: Easy scaling drives 20-30% annual upsells through natural usage growth
- **Risk Mitigation**: Customers can downgrade during slowdowns without penalty, reducing churn during economic downturns

Economic impact: Net Revenue Retention (NRR) of 110-130% driven by frictionless expansion—every $1 of new ARR becomes $1.10-$1.30 the following year.

12.3.4 4. Hybrid Pay-As-You-Go + Subscription Model

What it means: Customers buy bundled tiers (predictable base) + pay overage fees for usage spikes (consumption alignment).

Why this wins:

- **Land-and-Expand**: Start with small subscription, grow into higher tiers as usage increases organically
- **Revenue Upside**: Overage fees capture unexpected growth (20-30% of customers exceed base tier monthly)
- **Budget Certainty**: Base subscription is fixed; overage only applies when customers scale, aligning costs with their revenue growth
- **Self-Service Friendly**: Usage-based pricing removes sales friction—customers self-select appropriate tiers

Revenue mix: 60% subscription (predictable) + 30% infrastructure pass-through (annual) + 10% overage/services (expansion) = highly diversified revenue base.

12.3.5 5. Proprietary Ecosystem Lock-in

What it means: The Cosma Vector Database (CVD) is tightly coupled with our model architecture.

Why this wins:

- **Technical Dependency**: CVD's "zero-build" efficiency relies on specific vector properties produced only by Cosma models or our adaptation API.
- **No "Bring Your Own Vector" Leakage**: Customers cannot simply generate vectors with OpenAI and query CVD directly. They *must* use our adaptation layer, ensuring we capture revenue even if they use third-party models.
- **Sticky Ecosystem**: The database and model reinforce each other—using one makes the other more valuable.

12.3.6 6. Volume Expansion (The Jevons Paradox)

What it means: By reducing the unit cost of vector search by 10x, we unlock massive-scale use cases that were previously cost-prohibitive.

Why this wins:

- **New Workloads**: Instead of just indexing "hot" data (recent 10%), customers can now afford to index *everything* (archival data, logs, full history).
- **Higher Aggregate Spend**: While unit price is lower, the volume of indexed data explodes (100x), leading to significantly higher total contract values.
- **Data Gravity**: Once a customer indexes petabytes of data with us (because it's cheap), moving it becomes impractical, creating immense churn resistance.

Combined Strategic Impact

- **Customer Retention**: Annual contracts + zero-fee scaling + transparent pricing = ¡5% churn (vs. 10-15% industry average)
- **Net Revenue Retention**: Easy expansion + usage-based overage = 110-130% NRR (best-in-class SaaS metric)
- **Gross Margins**: 70% by Year 5 despite at-cost infrastructure (we monetize intelligence, not infrastructure)
- **Competitive Defensibility**: Pass-through pricing + decoupled architecture = nearly impossible for competitors to match without rebuilding their entire platform

12.4 COEM Model Pricing Tiers

COEM models are available in multiple dimensions, with pricing reflecting computational cost. Subscription tiers bundle prepaid COEM tokens with managed infrastructure and unlimited search queries.

Model	Dims	$/1M Tokens	Rel. Cost	Use Case
COEM-Tiny	128	$0.02	1x	High-volume, basic similarity
COEM-Small	256	$0.04	2x	Standard applications
COEM-Base	512	$0.08	4x	Production workloads
COEM-Large	768	$0.12	6x	High-accuracy search
COEM-XL	1024	$0.18	9x	Premium semantic understanding
COEM-XXL	1536	$0.30	15x	Maximum precision

Table 1: COEM Pricing by Dimension (100+ Languages, Unlimited Searches)

Pricing factors: Higher dimensions = more GPU memory, larger index storage, lower throughput per GPU, but better semantic precision.

Example: The "Professional" tier ($999/month) includes 5M embeddings (40B tokens using COEM-Base at $0.08/1M = $3.20 in compute) plus hosted infrastructure and unlimited searches.

12.5 Customer Segmentation

Segment	Typical Use Case	Infrastructure Tier	Monthly Embeddings	Price Range
Individuals/ Hobbyists	Personal projects, learning, prototyping	Shared CPU	10K-100K	Free/$9-29
Small Startups	MVP development, early apps with ¡10K users	1-2 vCPU instance	100K-1M	$99-299
Medium Companies	Production apps, 10K-100K users	4-8 vCPU instance	1M-10M	$999-2,999
Large Enterprises	Multiple products, 100K-1M users	Dedicated cluster (16-32 vCPU)	10M-100M	$9,999-29,999
Tech Giants	Platform-scale, millions of users	Multi-region clusters	100M+	Custom (6 figures)

Table 2: Customer Segment Characteristics (Unlimited Searches on Hosted Infrastructure)

12.6 Unit Economics by Segment

12.6.1 Individuals & Hobbyists

Revenue Model:

- Free Tier: Shared CPU + 10,000 embeddings/month (unlimited searches, $0 revenue)
- Starter: $9/month - Shared CPU + 50,000 embeddings (unlimited searches)
- Pro: $29/month - 1 vCPU + 200,000 embeddings (unlimited searches)
- **Included Model:** COEM-Small (256 dims, $0.04/1M tokens)
- Overage: $0.10 per 10K embeddings beyond quota (COEM-Small)

Cost Analysis (per paying user):

- Infrastructure (shared compute): $2-4/month
- Model inference (GPU, batched): $1-3/month
- Storage & bandwidth: $0.50-2/month
- **Total COGS:** $3.50-9/month
- **Gross margin:** 0-70% (Free tier subsidized by paid users)

Conversion assumptions: 5-10% of free users convert to paid tiers.

12.6.2 Small Startups

Revenue Model:

- Growth: $99/month - 1 vCPU instance + 500K embeddings (unlimited searches)
- Business: $299/month - 2 vCPU instance + 2M embeddings (unlimited searches)
- **Included Model:** COEM-Small (256 dims, $0.04/1M tokens)
- Overage: $0.10 per 10K embeddings beyond quota (COEM-Small)

Cost Analysis (per customer):

- Infrastructure (1-2 vCPU): $20-40/month
- Model inference (GPU): $10-25/month
- Storage & bandwidth: $5-10/month
- Support overhead: $5-10/month (email support)
- **Total COGS:** $40-85/month
- **Gross margin:** 60-75%

Revenue projections:

- Average deal size: $150-250/month (including overage)
- CAC: $500-1,000 (self-service, marketing-driven)
- Payback period: 4-6 months
- Annual LTV: $1,800-3,000 (assuming 12-18 month retention)

12.6.3 Medium Companies

Revenue Model:

- Professional: $999/month - 4 vCPU instance + 5M embeddings (unlimited searches)
- Premium: $2,999/month - 8 vCPU instance + 20M embeddings (unlimited searches)
- **Included Model:** COEM-Base (512 dims, $0.08/1M tokens)
- Overage: $0.12 per 10K embeddings (COEM-Base)
- Includes: 99.9% SLA, dedicated support, advanced features

Cost Analysis (per customer):

- Infrastructure (4-8 vCPU): $200-400/month
- Model inference (GPU): $100-250/month
- Storage & bandwidth: $50-100/month
- Support & account management: $100-150/month
- **Total COGS:** $450-900/month

- **Gross margin:** 65-75%

Revenue projections:

- Average deal size: $1,500-3,500/month
- CAC: $5,000-10,000 (direct sales)
- Payback period: 4-6 months
- Annual LTV: $30,000-50,000

12.6.4 Large Enterprises

Revenue Model:

- Enterprise: $9,999/month - 16 vCPU cluster + 50M embeddings (unlimited searches)
- Enterprise Plus: $29,999/month - 32 vCPU cluster + 200M embeddings (unlimited searches)
- **Included Model:** COEM-Large (768 dims, $0.12/1M tokens)
- Overage: $0.15 per 10K embeddings (COEM-Large)
- Includes: Single-tenant, VPC peering, premium support, compliance

Cost Analysis (per customer):

- Dedicated infrastructure (16-32 vCPU): $2,000-5,000/month
- GPU compute cluster: $1,500-3,500/month
- Storage & bandwidth: $500-1,000/month
- Premium support & CSM: $1,000-2,000/month
- **Total COGS:** $5,000-11,500/month
- **Gross margin:** 50-70%

Revenue projections:

- Average deal size: $15,000-35,000/month
- CAC: $30,000-60,000 (enterprise sales)
- Payback period: 6-12 months
- Annual LTV: $200,000-450,000
- Contract length: 1-3 years

12.6.5 Technology Giants

Revenue Model:

- Custom pricing: $100,000-500,000+/month
- **Included Model:** COEM-XL or COEM-XXL (1024-1536 dims, negotiated)
- Overage: Custom volume-based pricing
- Volume discounts, custom SLAs, white-glove service
- Platform integration, co-marketing opportunities

Cost Analysis (per customer):

- Massive dedicated infrastructure: $30,000-100,000/month
- GPU fleets: $20,000-80,000/month
- Storage & global CDN: $5,000-20,000/month
- Dedicated engineering & support: $10,000-25,000/month

- **Total COGS:** \$65,000-225,000/month
- **Gross margin:** 35-55%

Revenue projections:

- Average deal size: \$150,000-400,000/month
- CAC: \$100,000-250,000 (strategic sales)
- Payback period: 12-18 months
- Annual LTV: \$2M-5M+
- Contract length: 2-5 years

12.7 Revenue Stream Mix

Our revenue model evolves as the business scales, with infrastructure and services revenue growing alongside core model usage:

Revenue Stream	Years 1-2	Years 3-5
Model Usage Fees	60%	55%
Managed Infrastructure	30%	35%
Professional Services	10%	10%

Table 3: Projected Revenue Mix by Stream

Infrastructure revenue grows as customers scale to larger deployment tiers, while professional services remain stable at 10% as improved documentation and self-service tooling reduce dependency.

12.8 5-Year Revenue Projection

Segment	Year 1	Year 2	Year 3	Year 4	Year 5
Individuals (paid)	\$50K	\$200K	\$500K	\$1.0M	\$1.5M
Small Startups	\$300K	\$1.2M	\$3.0M	\$6.0M	\$10M
Medium Companies	\$500K	\$2.5M	\$7.5M	\$15M	\$25M
Large Enterprises	\$400K	\$3.0M	\$10M	\$25M	\$50M
Tech Giants	\$0	\$1.8M	\$7.2M	\$18M	\$36M
Total ARR	**\$1.25M**	**\$8.7M**	**\$28.2M**	**\$65M**	**\$122.5M**

Table 4: Projected Annual Recurring Revenue by Segment

12.9 Key Growth Assumptions

1. Customer acquisition ramps progressively: 200 → 800 → 2,500 → 5,000 → 8,000 total customers
2. Average revenue per customer grows from \$520/month (Y1) to \$1,275/month (Y5)
3. Gross margin improves from 55% (Y1) to 70% (Y5) through scale efficiencies
4. **Churn rates:** 5%/month (hobbyists), 3%/month (SMB), 1%/month (enterprise), 0.5%/month (tech giants)
5. **Net Revenue Retention:** 110-130% driven by usage growth and tier upgrades
6. Infrastructure costs decrease 15-20% annually through optimization and better pricing
7. Sales efficiency improves 30% year-over-year as product matures

12.10 Lean Team Strategy: Capital-Efficient Scaling

Unlike traditional SaaS companies that aggressively hire large sales and support teams, Azetta.ai is designed for **extreme operational leverage** through automation, product-led growth, and technical excellence.

12.10.1 Headcount Projection

Function	Year 1	Year 2	Year 3	Year 4	Year 5
Engineering (Product & Infra)	3	5	8	12	15
Sales & Marketing	1	2	4	6	8
Customer Success & Support	0.5	1	2	4	6
G&A (Finance, Legal, Ops)	0.5	1	2	3	6
Total Headcount	**5**	**9**	**16**	**25**	**35**
Revenue per Employee	**$250K**	**$967K**	**$1.76M**	**$2.6M**	**$3.5M**

Table 5: Projected Headcount & Revenue per Employee

12.10.2 How We Stay Lean

1. **Product-Led Growth (PLG)**: Self-service onboarding and freemium tiers eliminate need for large SDR/BDR teams. Target: 70%+ customers acquired through product, not outbound sales
2. **Automated Customer Success**: Built-in usage analytics, proactive alerts, and comprehensive documentation reduce support tickets by 60%. One CSM can manage 100+ accounts vs. industry standard of 30-40
3. **Engineering Leverage**: Investing in infrastructure automation (auto-scaling, monitoring, deployment) means 15 engineers in Year 5 can support $122M ARR workloads that competitors need 50+ engineers to manage
4. **Founder-Led Sales (Early Stage)**: Founders handle enterprise deals through Year 2, deferring expensive sales hires until repeatable playbook is proven
5. **Outsourced Non-Core Functions**: Accounting, legal, HR, and recruiting handled via fractional executives and external vendors rather than full-time hires

12.10.3 Benchmark Comparison

Metric	Azetta.ai	Typical SaaS	Advantage
Revenue per Employee (Year 5)	$3.5M	$150K-250K	**10-20x**
Customer Support Ratio	1:100+	1:30-40	**3x leverage**
Sales Team % of Headcount	23%	40-50%	**50% reduction**
Eng % of Headcount	43%	25-30%	**Product focus**

Table 6: Lean Operations vs. Industry Benchmarks

Capital Efficiency Impact: By Year 5, Azetta.ai achieves $3.5M revenue per employee—**10-20x higher than typical SaaS companies**. This operational leverage is the primary driver of our 18% EBITDA margins and positions us to reach profitability faster with less dilution.

12.11 Path to Profitability

Break-even: EBITDA-positive by Year 4 at $65M ARR. Company prioritizes aggressive market share capture in Years 1-3 while steadily improving operational efficiency. By Year 5, achieves

Metric	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Total Revenue	$1.25M	$8.7M	$28.2M	$65M	$122.5M
COGS (45-30%)	-$563K	-$2.96M	-$9.02M	-$19.5M	-$36.75M
Gross Profit	**$687K**	**$5.74M**	**$19.18M**	**$45.5M**	**$85.75M**
Gross Margin	**55%**	**66%**	**68%**	**70%**	**70%**
Sales & Marketing	-$875K	-$4.35M	-$9.87M	-$16.25M	-$24.5M
(as % revenue)	(70%)	(50%)	(35%)	(25%)	(20%)
R&D	-$500K	-$3.05M	-$8.46M	-$16.25M	-$24.5M
(as % revenue)	(40%)	(35%)	(30%)	(25%)	(20%)
G&A	-$188K	-$1.04M	-$3.38M	-$7.8M	-$14.7M
(as % revenue)	(15%)	(12%)	(12%)	(12%)	(12%)
EBITDA	**-$876K**	**-$2.7M**	**-$2.53M**	**+$5.2M**	**+$22.05M**
EBITDA Margin	**-70%**	**-31%**	**-9%**	**+8%**	**+18%**

Table 7: Projected Income Statement & Path to Profitability

best-in-class SaaS metrics with 70% gross margins and 18% EBITDA margins, demonstrating sustainable unit economics at scale.

Key efficiency drivers:

- **S&M optimization:** Product-led growth (PLG) motion reduces CAC by 60% from Year 1 to Year 5; self-service adoption drives 70%+ of new customers by Year 5
- **R&D leverage:** Core platform matures by Year 3; later-stage R&D focuses on incremental features and vertical expansion rather than foundational rebuilds
- **Infrastructure efficiency:** Economies of scale reduce per-customer compute costs 40% through better GPU utilization and AWS volume discounts

Note: This trajectory balances growth and profitability. More aggressive growth scenario (maintaining 30% S&M spend through Year 5) could achieve $200M+ ARR but delay profitability to Year 6.

13 Intellectual Property

13.1 Proprietary Technology

- **CVD Search Algorithm**: Trade secret - deterministic vector traversal method
- **CosmaDistance Metric**: Proprietary distance formulation optimized for cross-lingual search
- **COEM Training Pipeline**: Trade secret - alignment methodology and loss functions
- **Quantization Methods**: Trade secret - vector compression with minimal quality loss

13.2 Licensing Strategy

- Closed-source core technology
- Open-source client SDKs (community building)
- API-first distribution (SaaS default)
- Enterprise source licensing (restricted redistribution)

14 Risk Factors & Mitigation

As with any early-stage technology company, Azetta.ai faces several categories of risk. We have identified and assessed these risks and developed mitigation strategies for each.

14.1 Market & Competitive Risks

14.1.1 Risk: Competitive Replication

Description: Competitors or well-funded entrants could attempt to replicate our technology, potentially commoditizing our offerings.

Mitigation:

- **Trade Secret Strategy**: Our core innovations (CVD, COEM, CSL) are maintained as proprietary trade secrets, never published or patented. This prevents disclosure while maintaining exclusive use.
- **API-Only Access**: Technology is accessible only through our API—no open-source components, no model downloads, no architectural disclosure
- **Zero Build Time Moat**: CVD's instant document insertion (vs HNSW's hours-long index rebuilds) is an architectural advantage competitors cannot easily replicate without ground-up redesign
- **Operational Moat**: The complexity of our deployment, optimization, and operational expertise creates a significant barrier to replication. Even if competitors understand the concept, execution at scale requires years of specialized knowledge.
- **Implementation Complexity**: Our system requires deep expertise across vector databases, multilingual NLP, distributed systems, and kernel methods—a rare combination
- **First-Mover Advantage**: Building deep integration partnerships and customer lock-in while others are still developing
- **Network Effects**: Our multi-tenancy architecture creates data advantages that improve with scale
- **Continuous Innovation**: Ongoing R&D keeps us ahead of potential copycats—by the time someone replicates our current system, we're already two generations ahead

14.1.2 Risk: Big Tech Competition

Description: Google, Microsoft, or Amazon could enter the omnilingual semantic search space with significant resources.

Mitigation:

- **Specialized Focus**: Unlike general-purpose AI providers, we are laser-focused on omnilingual search optimization
- **Cost Efficiency**: Our architecture is fundamentally more cost-efficient than transformer-based approaches at scale

- **Partnership Potential**: We complement rather than compete with cloud providers; potential acquisition target
- **Nimble Execution**: As a startup, we can iterate and deploy faster than large organizations
- **Customer Intimacy**: Deep integration with early customers creates switching costs

14.1.3 Risk: Market Timing

Description: The market for omnilingual semantic search may not develop as quickly as projected.

Mitigation:

- **Early Customer Validation**: Focusing on design partnerships and LOIs to validate demand
- **Horizontal Applicability**: Our technology works across industries, reducing dependence on any single vertical
- **Cost Savings Appeal**: Even without growth, our 60% cost reduction drives adoption in existing search workloads
- **Pivot Flexibility**: Core technology applicable to other semantic understanding use cases

14.2 Technical Risks

14.2.1 Risk: Scaling Challenges

Description: Performance or reliability issues could emerge at enterprise scale.

Mitigation:

- **Proven Architecture**: CVD and COEM designs tested on large-scale datasets
- **Gradual Rollout**: Phased deployment starting with smaller customers
- **Multi-Cloud Strategy**: Redundancy across AWS, GCP, and Azure
- **Observability**: Comprehensive monitoring and alerting infrastructure from day one
- **Expert Team**: Deep distributed systems and ML infrastructure expertise on founding team

14.2.2 Risk: Model Performance

Description: Our models may not deliver expected accuracy or language coverage in production.

Mitigation:

- **Benchmark Validation**: Extensive testing on standard datasets before production

- **Continuous Improvement**: Active learning pipelines to improve performance over time
- **Hybrid Approach**: Fallback to established embedding models when needed
- **Transparent Metrics**: Clear communication of performance tradeoffs with customers

14.3 Execution Risks

14.3.1 Risk: Customer Acquisition

Description: Longer-than-expected sales cycles or high CAC could impact runway.

Mitigation:

- **PLG Motion**: Self-serve tier reduces dependency on enterprise sales
- **Design Partners**: Co-development approach with early customers
- **Content Marketing**: Technical thought leadership to drive inbound interest
- **Partner Channels**: Leverage system integrators and cloud marketplaces
- **Capital Efficient**: 18-month runway provides time to iterate go-to-market

14.3.2 Risk: Operational Complexity

Description: Managing multi-cloud infrastructure and customer deployments is operationally intensive.

Mitigation:

- **Automation First**: Heavy investment in IaC and deployment automation
- **Standard Configurations**: Limited SKUs reduce operational burden
- **Managed Services Tier**: SaaS option for customers who want hands-off
- **Observability**: Proactive monitoring reduces reactive firefighting

14.4 Regulatory & Compliance Risks

14.4.1 Risk: Data Privacy Regulations

Description: GDPR, CCPA, and emerging AI regulations could impact operations.

Mitigation:

- **Privacy by Design**: Data minimization and encryption built in from day one
- **Regional Deployment**: Data residency options for EU, US, and APAC
- **Compliance Roadmap**: SOC 2 Type II and ISO 27001 on 12-month roadmap
- **Customer Control**: Customers retain full control over their data

14.4.2 Risk: AI Liability & Ethics

Description: Emerging regulations around AI bias, explainability, and liability.

Mitigation:

- **Transparent Models**: CVD design is more interpretable than blackbox transformers
- **Bias Testing**: Regular fairness audits across languages and demographics
- **Customer Responsibility**: Clear ToS defining customer responsibility for use cases

14.5 Financial Risks

14.5.1 Risk: Longer Runway Needed

Description: Achieving product-market fit may take longer than 18 months, requiring additional capital.

Mitigation:

- **Milestone-Based Approach**: Clear go/no-go decisions at 6, 12, and 18 months
- **Capital Efficiency**: cloud-based development minimize burn
- **Revenue Acceleration**: Self-serve PLG tier can generate early revenue
- **Strong Investor Network**: Founders have access to follow-on capital if needed
- **Bridge Options**: Strategic partnerships could provide non-dilutive funding

14.6 Risk Management Philosophy

Our approach to risk management is proactive and transparent:

1. **Regular Assessment**: Monthly risk reviews with board/advisors
2. **Scenario Planning**: Best case, base case, worst case planning for all key decisions
3. **Transparent Communication**: Honest dialogue with investors about challenges
4. **Rapid Iteration**: Quick pivots when data suggests strategy adjustment needed
5. **Focus on What We Can Control**: Execution excellence and customer success over market speculation

While these risks are real, our team's experience, technical differentiation, and capital-efficient approach position us well to navigate the challenges of building a category-defining company in the semantic search space.

15 The Ask: $1.11M Pre-Seed Round

15.1 Funding Request

Azetta.ai is raising **$1.11 million** in pre-seed funding to build and validate Cosma as the leading omnilingual semantic search platform. This capital will provide an **18-month runway** to achieve product-market fit and position the company for a successful Series A.

15.2 Use of Funds

The funding allocation is designed to balance aggressive product development with disciplined capital efficiency:

Category	Item	Amount
Personnel (59% of budget)		
Core Team Salaries	CEO (Taha Bouhsine) — 18 months	$120,000
	CTO (Douglas Seo) — 18 months	$120,000
	COO (Poppy Yuan) — 18 months	$120,000
Engineering Hires	ML Engineer (salary & benefits) — 18 months	$165,000
	Backend Engineer (salary & benefits) — 15 months	$135,000
Subtotal		**$660,000**
Research & Compute (27% of budget)		
Model Training	Large-scale multilingual model training (cloud GPU clusters)	$250,000
Production Infrastructure	API hosting, inference serving, and monitoring	$35,000
Data Acquisition	Multilingual training datasets and preprocessing	$15,000
Subtotal		**$300,000**
General & Administrative (14% of budget)		
Office & Operations	Co-working space ($3K/month × 18 months)	$54,000
Professional Services	Legal (incorporation, IP), accounting, compliance	$40,000
Software & Tooling	GitHub, Slack, monitoring, analytics, productivity tools	$18,000
Marketing & Sales	Conference attendance, early customer outreach	$20,000
Contingency	Reserve for unforeseen expenses	$18,000
Subtotal		**$150,000**
Total Funding Request		**$1,110,000**

15.3 Key Milestones with This Funding

The $1.11M will enable Azetta.ai to achieve the following critical milestones over 18 months:

15.3.1 Technical Milestones (Months 1-9)

- **Q1 (Months 1-3)**: Production-ready CVD v1.0 with deterministic sub-50ms retrieval across 10M+ vectors

- **Q2 (Months 4-6)**: COEM multilingual models trained on 50+ languages with SOTA cross-lingual transfer performance
- **Q3 (Months 7-9)**: Cosma Search Layer (CSL) deployed with auto-scaling, multi-tenancy, and enterprise security (SOC 2 Type I)

15.3.2 Commercial Milestones (Months 6-18)

- **Q2-Q3 (Months 6-9)**: 5-10 design partner deployments across target verticals (e-commerce, SaaS, financial services); establish product-market fit signals
- **Q4 (Months 10-12)**: Launch commercial tiers with first 10-15 paying customers; achieve $300K-400K ARR run rate
- **Month 12 (End of Year 1)**: Reach $1.2M-1.5M ARR with 150-250 total customers (mix of hobbyist, startup, and enterprise tiers)
- **Months 13-18**: Scale to $5M-7M ARR run rate with 500-800 customers; refine enterprise playbook and establish repeatable sales motion

These milestones are ambitious but achievable because of how we've architected both our product *and* our business model for capital efficiency.

15.4 Capital Efficiency Strategy

Azetta.ai is optimizing burn rate through:

- **Lean Core Team**: Founders take below-market salaries ($80K/year vs. $150K+ market rate) to extend runway
- **Strategic Compute Spending**: Leverage spot instances and government research grants (Taha's NSERC connections) to reduce GPU costs by 40-60%
- **Design Partner Model**: Early customers provide feedback and validation in exchange for discounted pricing, reducing CAC to near-zero for first 10 customers
- **Open-Source Leverage**: Release CVD core as open-source to drive developer adoption and community validation, reducing marketing costs

Beyond these operational efficiencies, our business model itself is designed to create structural advantages that reduce risk and accelerate growth.

15.5 Business Model Moats: Built-In Defensibility

Beyond product differentiation, Azetta.ai's pricing architecture creates **structural competitive advantages** that de-risk this investment:

15.5.1 1. Self-Funding Growth Engine

Annual infrastructure commitments mean customers prepay 12 months of cloud costs upfront:

- **Year 1**: $375K in prepaid cash (30% of $1.25M ARR)
- **Year 2**: $2.6M in prepaid cash (30% of $8.7M ARR)
- **Year 3**: $8.5M in prepaid cash (30% of $28.2M ARR)

Investor impact: By Year 3, we're generating **$8.5M+ in upfront customer cash** annually—enough to self-fund operations with minimal equity dilution. This reduces Series B dependency and improves founder/investor alignment.

15.5.2 2. Competitive Immunity from At-Cost Pass-Through

Our **zero-markup infrastructure pricing** creates an impossible-to-beat moat:

- Competitors charge 20-40% markup on infrastructure ($6K-7K for workloads we charge $5K for)
- **They cannot undercut us**—matching our pricing destroys their margins
- We profit from high-margin intelligence (COEM models), not commodity infrastructure
- Enterprise buyers value radical transparency—builds trust and long-term partnerships

Defensive positioning: Competitors must either (1) match our transparency and lose margin, or (2) maintain markups and lose customers. We win either way.

15.5.3 3. Expansion Revenue Built Into Architecture

Zero-fee scaling (enabled by decoupled storage) drives organic expansion:

- Customers upgrade **without penalty**—eliminates friction that kills upsells at competitors
- Target: **110-130% Net Revenue Retention (NRR)** by Year 3
- 20-30% of customers pay usage overage monthly (automatic expansion revenue)
- Can downgrade during downturns without penalty—reduces churn risk to ¡5%

Math: Every $1 of new ARR becomes $1.10-$1.30 the following year through natural expansion. This compounds: $1M ARR → $1.1M → $1.21M → $1.33M over 3 years *with zero additional sales effort*.

15.5.4 4. Technical Moat = Pricing Moat

Our decoupled architecture isn't just a technical feature—it's a **pricing differentiator competitors can't match**:

- Traditional vector databases couple storage + compute, forcing customers into rigid instance tiers
- We decouple storage, enabling zero-fee scaling up/down
- **Competitors would need 12-18 months to rebuild their architecture** to match this capability
- First-mover advantage locks in customers before competitors can respond

Why These Moats Matter for Investors

- **Predictable Revenue**: Annual contracts + ¡5% churn = highly forecastable ARR growth
- **Capital Efficiency**: $8.5M+ prepaid cash by Year 3 = reduced reliance on equity financing
- **Defensive Positioning**: Competitors can't match pricing without rebuilding platforms or destroying margins
- **Compounding Growth**: 130% NRR means revenue compounds faster than sales investment required
- **Exit Attractiveness**: Strategic acquirers (AWS, Google Cloud, Databricks) see ecosystem control value in owning both infrastructure + intelligence layer

This combination—ambitious milestones, operational discipline, structural business model advantages, and risk mitigation—positions us to achieve exactly what institutional VCs need to see for Series A.

15.6 Why This Round Sets Us Up for Series A

This pre-seed round is designed to achieve the *minimum viable traction* institutional VCs require for Series A:

Series A Requirements: What We'll Deliver

- **Revenue Traction**: $5M-8M ARR by Month 18-24 (exceeds typical $3M-5M Series A threshold)
- **Product-Market Fit Signals**: 500-800 customers across all tiers, ¡5% monthly churn, NPS ¿50
- **Technical Differentiation**: Proven 10-100x cost advantage vs. competitors in 50+ production deployments
- **Repeatable GTM**: Documented sales playbook with predictable CAC/LTV ratios and established conversion funnels
- **Team Strength**: Engineering team scaled to 7-10 people with demonstrated ability to ship complex infrastructure at scale

15.7 Post-Series A Vision (18-36 Months)

With Series A capital ($5-8M), Azetta.ai will:

1. **Scale GTM**: Build dedicated sales team (3-5 AEs) and expand to 100+ enterprise customers ($5M ARR)
2. **Product Expansion**: Add vertical-specific features (e-commerce, legal tech) and enterprise integrations (Salesforce, ServiceNow)
3. **Global Expansion**: Establish European presence for GDPR-compliant deployments and multilingual support
4. **R&D Investment**: Expand COEM to 100+ languages and build advanced features (RAG, multi-modal search)

This clear progression—$1.11M pre-seed to prove product-market fit, Series A to scale GTM, Series B to dominate market—gives investors confidence in a predictable path to a category-defining company.

Bottom line: The $1.11M pre-seed is not about achieving profitability—it's about proving that Cosma solves a painful, expensive problem better than anyone else, with a clear path to $25M+ ARR by Year 3 and positioning for a strong Series B at $50M+ ARR.

16 Conclusion

Azetta.ai's **Cosma** platform represents a fundamental advance in semantic search technology. By combining proprietary vector database infrastructure with unified omnilingual embeddings, we enable organizations to search across languages with unprecedented accuracy and performance.

Investment Highlights

- **Market Size**: $8B+ global semantic search market, 35% CAGR
- **Technology Moat**: Trade secret algorithms, proprietary models
- **Enterprise-Ready**: Security, compliance, dedicated cloud
- **Scalable Economics**: High-margin SaaS + usage-based pricing
- **Proven Traction**: [Add customer metrics, revenue, growth]

Contact Information
Azetta.ai

This document contains confidential and proprietary information. Distribution is restricted to qualified investors under NDA.

A Competitor Pricing Details

A.1 Voyage AI Detailed Pricing

A.1.1 Text Embedding Models

Model	Price (per 1M tokens)	Free Tier
voyage-3-large	$0.18	200M tokens
voyage-context-3	$0.18	200M tokens
voyage-3.5	$0.06	200M tokens
voyage-3.5-lite	$0.02	200M tokens
voyage-code-3	$0.18	200M tokens
voyage-finance-2	$0.12	50M tokens
voyage-law-2	$0.12	50M tokens
voyage-code-2	$0.12	50M tokens

Table 8: Voyage AI Text Embedding Pricing

A.1.2 Multimodal Embeddings

- **Model**: voyage-multimodal-3
- **Text Price**: $0.12 per 1M tokens
- **Image Price**: $0.60 per 1B pixels
- **Free Tier**: 200M text tokens + 150B pixels
- **Image Cost Constraints**: Min $0.00003 (small images), Max $0.0012 (large images)

A.1.3 Rerankers

Model	Price (per 1M tokens)	Free Tier
rerank-2.5	$0.05	200M tokens
rerank-2.5-lite	$0.02	200M tokens

Table 9: Voyage AI Reranker Pricing

A.2 Vector Scaling Costs

- **Voyage AI**:
  - *Scaling Model*: No direct storage costs (embedding provider only).
  - *Cost Factor*: Costs scale linearly with embedding generation volume (tokens processed).
  - *Optimization*: Offers Matryoshka embeddings (256-2048 dims) and quantization (int8/binary) to reduce downstream storage costs by up to 6x.
- **Weaviate (Serverless)**:
  - *Scaling Model*: Charged per 1 million stored dimensions per month.
  - *Standard Tier*: $\approx$$0.095 per 1M dimensions/month (starts at $25/mo).
  - *Business Critical*: $\approx$$0.175 per 1M dimensions/month.
  - *Calculation*: Total Cost = (Number of Vectors $\times$ Dimensions per Vector) $\times$ Rate.
- **Pinecone (Serverless)**:

  - *Scaling Model*: Charged per GB of stored data per month.
  - *Storage Rate*: $0.33 per GB/month.
  - *Includes*: Vector data (dense/sparse), metadata, and IDs.
  - *Note*: Separate costs for Read/Write units ($8.25 per 1M units).

- **pgvector (Self-Hosted/AWS RDS)**:

  - *Scaling Model*: Infrastructure-based (Storage + RAM).
  - *Memory Requirement*: High. HNSW index typically requires 2-3x the raw vector size in RAM for optimal performance.
  - *Scaling Cost*: Step-function cost increases when upgrading to larger RAM instances (e.g., AWS r6g.xlarge to 2xlarge).
  - *Optimization*: `halfvec` (2-byte floats) and quantization can reduce RAM needs by 50%+.

A.3 Search Cost Scaling (1M vs 100M vs 1B Vectors)

- **Voyage AI**:

  - *Search Cost*: **$0** (External). Voyage charges only for embedding the query text.
  - *Scaling*: Constant cost per query, regardless of database size (1M or 1B).

- **Weaviate (Serverless)**:

  - *1M Vectors*: Covered by Standard Tier (≈$25/mo).
  - *100M Vectors*: Cost is driven by **storage** (stored dimensions), not query volume. High monthly storage fee (≈$1,500+ for 1536d), but unlimited queries included.
  - *1B Vectors*: Requires Enterprise/Business Critical tier. Primary cost driver is the massive storage footprint, not the search operation itself.

- **Pinecone (Serverless)**:

  - *Query Cost*: Measured in Read Units (RUs).
  - *Scaling Behavior*: **Sublinear**. Querying a 4x larger index does *not* cost 4x RUs (e.g., 50k vectors ≈ 5 RUs, 200k vectors ≈ 8 RUs).
  - *1B Vectors*: While per-query RU efficiency improves, the sheer volume of data requires dedicated read nodes or high shard counts to maintain latency, increasing base infrastructure costs.

- **pgvector (Self-Hosted)**:

  - *1M Vectors*: Low cost. Fits in standard RAM (e.g., 4GB instance).
  - *100M Vectors*: **The RAM Cliff**. HNSW index for 1536d vectors requires ≈600GB RAM. Requires expensive high-memory instances (e.g., AWS r6g.24xlarge).
  - *1B Vectors*: **Prohibitive without Quantization**. Uncompressed HNSW requires ≈6TB RAM. Practical implementation mandates binary quantization or disk-based indexing (e.g., `pgvectorscale`) to avoid $20k+/month infrastructure bills.